NAME

mbpy_xyplot − Create an executable shellscript which will generate a GMT modern-mode plot of xy data from one or more data files.

VERSION

Version 5.0

SYNOPSIS

mbpy_xyplot −I*file*[**:***key=value...*] [**-I***file*[**:***key=value...*] ...] [**−O***root*] [**−−format***fmt*] [**−−title***text*] [**−−xlabel***text*] [**−−ylabel***text*] [**−−dpi***dpi*] [**−X**] [**−V**]
[**−R***xmin/xmax/ymin/ymax*[*r*]] [**−J***spec*] [**−P***pagesize*] [**−−portrait** | **−−landscape**] [**−B***spec*] [**−−length−scale***spec*]
[**−S***symbol*] [**−G***color*] [**−W***pen*]
[**−−coast**] [**−−coast−resolution***{f,h,i,l,c}*] [**−−coast−dry−fill***color*] [**−−coast−wet−fill***color*]
[**−−coast−lake−fill***color*] [**−−coast−river***color*] [**−−coast−boundary***spec*] [**−−coast−pen***pen*]
[**−−text***lon/lat/size/angle/font/justify/text*]
[**−−gmt−default***param/value*]

DESCRIPTION

mbpy_xyplot is a Python/PyGMT-era replacement for the Perl macro **mbm_xyplot**. It reads one or more xy data files and generates an executable shellscript of **GMT modern mode** commands (**gmt begin** / **gmt end**) which, when run, plots them as lines, symbols, or both, optionally overlaid with a coastline and/or text labels. Axes may be an auto-fit linear (Cartesian) scale or any GMT map projection.

The plot is scaled to fit the specified page size using the same page-layout, projection-fitting, and tick-interval heuristics as **mbm_histplot** -- confirmed by direct comparison of the two Perl macros' own sources to be identical (aside from whitespace), and distinct from **mbm_grdplot**'s own versions of the same computations (see **mbpy_histplot**(1)). In particular, when no **−−projection** is given, **mbpy_xyplot** computes two *independent* x and y scales that each separately stretch to fill the page (unlike **mbpy_grdplot**/**mbpy_grd3dplot**'s single aspect-ratio-preserving scale, which would be wrong here since an xy plot's x and y axes routinely have unrelated units and ranges -- e.g. a time-series plot).

Unlike **mbm_xyplot**'s single-letter **−M***sub***:***sub***:**... grammar for coastline/text-label/length-scale tuning, **mbpy_xyplot** exposes each of those as its own descriptive long option (see OPTIONS below).

mbpy_xyplot never writes a persistent temporary file, unlike **mbm_xyplot** itself: for every input file, **mbm_xyplot** preprocesses (delimiter-splits and, if requested, applies a column- selection/math expression to) the data into a file named <*input*><*n*>.<*pid*> left behind in the current directory after every run, whether or not the plot script is ever executed. **mbpy_xyplot** instead performs the equivalent column-selection/math once in memory, in Python, to determine the auto-fit plot bounds, and at plot time pipes each input file through a small **awk**(1) program straight into **gmt plot**'s standard input -- a live, ephemeral pipe, with nothing written to disk.

Each **−−input** file may carry its own colon-separated overrides, in *key=value* form: **symbol=**, **fill=**, **pen=** (styling, defaulting from **−S**/**−G**/**−W**), **delimiter=** (default: whitespace), and **xexpr=**/**yexpr=** (column-selection/math expressions using **mbm_xyplot**'s own **c[***N***]** field-selection and **#** line-number mini-language, e.g. **c[3]** or **c[3]*2+1**; default **xexpr=c[1]**, **yexpr=c[2]**). A line beginning with **>** is always passed through unchanged as a multi-segment marker, matching **mbm_xyplot**'s (and **gmt plot**'s) own convention.

A genuine bug was found and fixed while validating this port against real-world data: **mbm_xyplot**'s own per-line numeric-result check is unintentionally a no-op (its validation regex is satisfied by the empty string), so an ordinary text header line at the top of a data file passes through into the plotted file unfiltered -- and because

gmt plot infers each column's data type from the *first* record it reads, a single such non-numeric first line makes it silently discard every subsequent, perfectly valid row instead of just that one line. **mbpy_xyplot** validates each computed x/y pair with a real numeric check instead, skipping (with a warning) only the genuinely bad lines.

A related, deliberate addition with no equivalent in **mbm_xyplot** at all: a data column that is exactly constant (e.g. a real "heave" trace logged as 0.0 for an entire survey) collapses to a zero-width axis that neither classic nor modern GMT can project. **mbpy_xyplot** pads such a degenerate axis by a small margin so it still plots as a visible flat line instead of aborting.

mbpy_xyplot requires **GMT** 6 or later (modern mode), a POSIX **awk**(1), and a Python 3 interpreter. It depends on **mbpy_layout.py** and **baseline_gmt_defaults()** from **mbpy_grdplot_script.py** (shared with **mbpy_grdplot**/ **mbpy_grd3dplot**), and reuses the page-size/tick-interval helper functions from **mbpy_histplot_script.py**, plus its own **mbpy_xyplot_script.py**, all installed alongside it.

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OPTIONS

−I, **−−input** *file*[**:***key=value...*]
An xy data file to plot, with optional colon-separated overrides **symbol=**, **fill=**, **pen=**, **delimiter=**, **xexpr=**, **yexpr=** (see DESCRIPTION). Repeatable; required at least once.

−O, **−−output** *root*
Output root name. Default: the first **−−input** file's basename with its extension removed.

−−format *fmt*
Output format passed to **gmt figure**: **pdf**, **png**, **ps**, **jpg**, **tif**, **eps**, ... Default **pdf**.

−−title *text*
Plot title, placed above the frame.

−−xlabel *text***, −−ylabel** *text*
Axis labels. Only used for a linear/Cartesian projection (an auto-fit default, or an explicit **−Jx**/**−JX**); a geographic or other map projection has no per-axis label, matching **mbm_xyplot**.

−−dpi *dpi*
The final raster's resolution (via **gmt figure**).

−X, **−−execute**
Run the generated script immediately after writing it.

−V, **−−verbose**
Print the generated script's path and a one-line summary.

−R, **−−region** *xmin/xmax/ymin/ymax***[***r***]**
Plot region. Fields may be given as *dd:mm:ss*. A trailing **r** means the four values are two corner points (southwest, northeast) rather than a w/e/s/n range; a bare **r** reformats the auto-computed data bounds into that corner-point form. Default: computed from the data itself.

−J, **−−projection** *spec*
GMT map scale/projection spec, e.g. **x0.02** (linear, 0.02 inch/unit), **m1.0** (Mercator, 1 inch/degree). Default: an auto-fit linear projection with independent x and y scales (see DESCRIPTION), chosen to fill the page.

−P, **−−pagesize** *size*
Page size code: **a**−**f**, **e1** (US sizes), **a0**−**a10**, **b0**−**b10**, **c0**−**c7** (ISO/architectural), or **m1**−**m6** (large-format). Default **a**.

−−portrait, **−−landscape**
Force the page orientation instead of choosing automatically; when neither is given, this auto-fit independent-scale mode defaults to landscape, matching **mbm_xyplot**/**mbm_histplot**.

−B, **−−tick−info** *spec*
Raw **−B** basemap tick-interval spec, overriding the automatically computed one.

−−length−scale *spec*
A GMT distance (map) scale bar spec, passed as **−L***spec* on **gmt basemap**, e.g. **fx10/2/32.70/5.0+lkm** (**mbm_xyplot**'s **−MGL**). The literal word **center** appearing between two slashes is replaced with the plot's own data center (e.g. **fxcenter/5.0+lkm**), matching **mbm_xyplot**'s own substitution.

−S, **−−symbol** *symbol*
Default GMT symbol code and size for every **−−input** file that does not set its own **symbol=**, e.g. **c0.1** (circle, 0.1 inch). Default: none (points are connected with a line instead).

−G, **−−fill** *color*
Default symbol/polygon fill color for every **−−input** file that does not set its own **fill=**. Default: none.

−W, **−−pen** *pen*
Default line pen for every **−−input** file that does not set its own **pen=**, e.g. **1p,black**. Default: none.

−−coast
Enable a coastline overlay (**gmt coast**), drawn *after* the xy data plots and *before* text labels and the basemap frame, matching **mbm_xyplot**'s own layer order exactly (data points that fall on a solid land fill are covered by it, just as in the classic macro).

−−coast−resolution *{f,h,i,l,c}*
Coastline database resolution: full, high, intermediate, low, crude. Default **f** when **−−coast** is enabled with no other coastline option given.

−−coast−dry−fill, **−−coast−wet−fill**, **−−coast−lake−fill** *color*
Fill colors for land, water, and lakes. **−−coast−dry−fill** defaults to **200** (a mid-gray) when **−−coast** is enabled with no fill, pen, river, or boundary option given at all -- an **mbm_xyplot**-specific default with no equivalent in **mbm_grdplot**/**mbpy_grdplot**.

−−coast−river *color*
Draw rivers in the given color.

−−coast−boundary *spec*
Draw a national boundary; repeatable.

−−coast−pen *pen*
Pen for the coastline itself. Defaults to **1p** together with the **−−coast−dry−fill** default described above.

−−text *lon/lat/size/angle/font/justify/text*
Place a text label (**gmt text**), e.g. **−122.2/36.6/12/0/1/LB/Monterey Bay**. Repeatable.

−−gmt−default *param/value*
Override a GMT default parameter for this plot (e.g. **MAP_FRAME_TYPE/plain**); repeatable.

EXAMPLES

Plot a single time series of sensor depth values as a thin connecting line, rendered directly to a PNG:

```
mbpy_xyplot −I "depth_vs_time.txt:pen=0.5p,blue" −O depth_ts \
  −−title "Sensor Depth vs Time" −−xlabel "Time (s)" \
  −−ylabel "Sensor Depth (m)" \
  −−pagesize b −−landscape −−format png −−dpi 150 −−execute −V
```

Plot navigation fixes from a comma-delimited file (selecting the longitude and latitude columns explicitly) as symbols, with a coastline underlay:

```
mbpy_xyplot −I "fixes.csv:delimiter=,:xexpr=c[2]:yexpr=c[3]:symbol=c0.05:fill=red" \
  −O fixes_plot −−title "Vehicle Fixes" \
  −−coast −−coast−resolution i \
  −−format png −−execute −V
```

Plot two data files together, one as a line and one as symbols, over a user-specified region:

```
mbpy_xyplot −I "reference.xy:pen=1p,black" \
  −I "observed.xy:symbol=c0.08:fill=red" \
  −O comparison −R "−10/110/−5/55" \
  −−title "Reference vs Observed" −−format pdf −V
```

SEE ALSO

mbm_xyplot(1), **mbpy_grdplot**(1), **mbpy_grd3dplot**(1), **mbpy_histplot**(1), **awk**(1), **mbdefaults**(1), **mb-system**(1)

BUGS

Please report any bugs to the MB-System project on GitHub (https://github.com/dwcaress/MB-System).